Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Practical

Practical – IV

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Aim: Problem Solving by Searching (Informed Search)

1).Scenario:
A medical test is used to detect a particular disease.
The probability that a randomly selected person has the disease is 1%.
If a person has the disease, the test returns positive with probability 99%.
If a person does not have the disease, the test still returns positive with probability 5% (false positive rate).
A person takes the test and receives a positive result.

Problem Statement:
Write a Python program to compute the probability that the person actually has the disease given that the test result is positive.

CODE:

```
P_disease = 0.01
P_no_disease = 1 - P_disease
P_positive_given_disease = 0.99
P_positive_given_no_disease = 0.05
print("=== Bayes' Rule Application ===\n")
print(f"P(Disease) = {P_disease}")
print(f"P(No Disease) = {P_no_disease}")
print(f"P(Positive | Disease) = {P_positive_given_disease}")
print(f"P(Positive | No Disease) = {P_positive_given_no_disease}")
# Step 1: Calculate P(Positive)
print("\nStep 1: Calculate P(Positive)")
P_positive = (P_positive_given_disease * P_disease) + \
        (P_positive_given_no_disease * P_no_disease)
print(f"P(Positive) = (0.99 × 0.01) + (0.05 × 0.99)")
print(f"P(Positive) = {0.99 * 0.01:.4f} + {0.05 * 0.99:.4f}")
print(f"P(Positive) = {P_positive:.4f}")
# Step 2: Apply Bayes' Theorem
print("\nStep 2: Apply Bayes' Theorem")
P_disease_given_positive = (P_positive_given_disease * P_disease) / P_positive
print("Formula:")
print("P(Disease | Positive) = [P(Positive | Disease) × P(Disease)] / P(Positive)")
print(f"\nP(Disease | Positive) = (0.99 × 0.01) / {P_positive:.4f}")
print(f"P(Disease | Positive) = {0.99 * 0.01:.4f} / {P_positive:.4f}")
print(f"P(Disease | Positive) = {P_disease_given_positive:.4f}")
print(f"\nFinal Answer = {P_disease_given_positive * 100:.2f}%")
```

OUTPUT:

```
P(Positive | Disease) = 0.99
P(Positive | No Disease) = 0.05

Step 1: Calculate P(Positive)
P(Positive) = (0.99 x 0.01) + (0.05 x 0.99)
P(Positive) = 0.0099 + 0.0495
P(Positive) = 0.0594

Step 2: Apply Bayes' Theorem
Formula:
P(Disease | Positive) = [P(Positive | Disease) x P(Disease)] / P(Positive)

P(Disease | Positive) = (0.99 x 0.01) / 0.0594
P(Disease | Positive) = 0.0099 / 0.0594
P(Disease | Positive) = 0.1667

Final Answer = 16.67%
```